

A Complex Namespace for the **rho** Calculus

Paired messages, two quotations, and an intertwining COMM

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Version 4 — a record of the core calculus

Abstract

This note records, in one place, the core of a variant of the **rho** calculus designed so that the namespace carries the structure of a complex scalar field rather than that of a free commutative monoid. Three changes are made to the term language and one to the reduction relation. Messages carry *pairs of sequences* of processes, so that a message is a formal difference, negation is structural, and the namespace is graded by rank. Names are quotations of such pairs, so that negation is an operation on *addresses* and not merely a decoration on sends; negation then lifts uniquely to processes, and requiring reduction to commute with that lifting is what selects the deposit convention of the communication rule. There are two quotation operators and, of necessity, two dropping operators, and receipts bind a pair of sequences of variables because negation acts on a receipt by exchanging them rather than by descending into it; the two quotations are read as a real and an imaginary sector, and a drop reads a name into the sector it asks for by *rotating* the name there, so that no context can read the sector off a name. Communication *intertwines* the sectors, so that a communication event is multiplication by the imaginary unit, and communication between participants of opposite polarity exchanges the components of the received pair, so that opposite polarity contributes the sign. The note states what has been settled, states the design forks that have not, and enumerates the obligations the design incurs. It also records the evaluation product against which the eventual pairing is defined, together with the pole and the residuation that give the dagger, taking the pole to be the zero process and the perp to be residuation to it, and it defines the two multiplicatives that the sequence structure makes available. It proves almost nothing. Its purpose is to fix notation and to make the open questions precise enough to be worked on.

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1 What this note is

1.1 Scope

The present document is a *record*, not an argument. It writes down the term language, the structural theory and the reduction relation of a variant calculus that has emerged from a sequence of design decisions, together with the reasons each piece is present. It does not develop the semantics that motivated the design; that semantics — an interpretation of the operations of quantum mechanics over a reflective process calculus, with processes as vectors, names as scalars, evaluated contexts as covectors and contexts as operators — is the subject of separate work, and is recalled here only far enough to explain why each feature was added.

Two things are deliberately excluded. The note gives no metatheory: no subject reduction, no congruence result, no decidability. And it gives no semantics: no interpretation map, no inner product, no dagger beyond the one line needed to say which operation the dagger is built from. Both are named as obligations in Section 11.

1.2 Status markers

Throughout, claims carry one of three markers.

SETTLED The design decision has been taken and the consequences worked through. It may still be wrong, but it is not open.

FORK Two or more readings are consistent with what has been settled, and the choice has not been made. The alternatives are stated.

OPEN An obligation the design incurs and does not discharge.

1.3 Revision note

Version 1 of this note had receipts bind a pair of name variables and stated the communication rule in two clauses, one per relative polarity. Both features are withdrawn. Negation on names lifts uniquely to processes (Proposition 10), the whole received pair is deposited into a single bound variable, and the polarity clause becomes a factor in one rule. The change is not cosmetic: requiring reduction to be equivariant for the lifted negation *selects* the deposit convention, and version 1's convention fails that test (Remark 31). The surviving pair sort occurs only on messages and inside quotations.

Version 3 makes three further changes. Messages, binders and quotations carry *sequences* rather than single processes, with rank as a grading (Definition 3); the two multiplicatives $\otimes$ and $\wp$ are defined and distinguished by whether the binders are shared (Section 7); and the cross cases of the dropping operators are settled by *rotation* rather than projection to zero (Definition 15), which restores unconditional reflection and discharges the observability obligation. Two consequences are recorded rather than assumed: slotwise parallel composition cannot be the addition (Proposition 43), so the sum in the evaluation product is external; and the sector table for the multiplicatives is the same table as for communication, so the fork over that table cannot be settled independently for the two (Remark 50).

Version 4 corrects a bug in version 2 which version 3 inherited. Receipts bind a *pair* of sequences of variables, one per polarity, and negation acts on a receipt by exchanging them. Version 2 had removed the second binder on the ground that it was derivable as $\ominus$ of the first; that argument is circular, since $\ominus$ is defined by recursion over names and has no clause for a variable, so with a single binder the lifting of negation is undefined on open terms — and every body under a binder is open. Remark 11 gives the argument, Remark 29 records what survives of the version 2 collapse, and the communication rule of Definition 28 is restated to deposit slotwise into both sequences.

1.4 Notational rules

The calculus is called the **rho** calculus and is never written with the Greek letter that follows pi; the name abbreviates *reflective higher order*, and the transliteration is the pun. Quotation is written @ and dropping *; corner quotes are not used.

The design discussions that produced this calculus wrote the pair constructor as an infix asterisk, $Q_p * Q_n$. That notation collides with the dropping operator, so pairs are written here with angle brackets, $\langle Q_p, Q_n \rangle$, and the reader translating from those discussions should read $\langle Q_p, Q_n \rangle$ wherever $Q_p * Q_n$ appears there.

2 Why the namespace is expanded

The calculus below is the minimal response to four defects, each of which is a defect of the *unexpanded* namespace. They are recalled because each feature of the design is answerable to one of them, and a reader who forgets the defects will find the features arbitrary.

The scalars have no additive inverses. In the ordinary rho calculus, structural congruence is associativity, commutativity and a unit for parallel composition. Consequently $\text{Proc}/\equiv$ is the free commutative monoid on the non-parallel components of processes, and Name , being $\text{Proc}/\equiv$ transported through quotation, is that same free commutative monoid. A free commutative monoid has no inverses and no cancellation. Whatever else it can express, it cannot express the vanishing of a sum of nonzero contributions. Since destructive interference is exactly such a vanishing, no coefficient structure carried by names can produce it.

The multiplicative and additive units collide. Name multiplication is $@P \cdot @Q := @(P \mid Q)$, whose unit is $@0$. Any addition on names built from summation also has $@0$ for its unit. The rig law $0 \cdot x = 0$ then fails, since $@0 \cdot x = x$. There is no absorbing element, and hence no rig.

There is no conjugation. Parallel composition is commutative, so any pairing built from it is symmetric rather than Hermitian, and there is no involution available on the coefficients to repair this. An involution must come from somewhere; the unexpanded calculus has none of order two acting on scalars.

Multiplication by a monomial is a shift, not a scaling. This is the sharpest of the four. In the free commutative monoid, multiplying a name by another name is injective and never decreases anything; it permutes basis elements rather than combining them. Any attempt to realise the imaginary unit as an ordinary name — for instance by adjoining a ground process 1 with $1 \mid 1 \mid 1 \mid 1 \equiv 0$, so that $@1$ has order four — therefore fails for a reason that is fatal rather than technical: $@R$ and $@(1 \mid R)$ are names of *different addresses*, so no receiver hears both and nothing superposes. Multiplication by i must move a message *between sectors at one address*, not between addresses. This is the requirement that forces the sector to be a coefficient attached to a name rather than a factor inside it, and it is the reason the design expands the namespace instead of enriching the process language.

Remark 1. The four defects are not independent. Repairing the first by group completion supplies the involution needed for the third, and separates the units demanded by the second. Only the fourth is a genuinely separate requirement, and it is the one that dictates *where* the new structure is placed.

3 Syntax

3.1 Sorts

The calculus has three sorts: processes, pairs and names. Pairs are the new sort. A pair is intended as a formal difference of processes, its first component positive and its second negative.

Pairs occur in two places only, on messages and inside quotations; binders do not mention them, for the reason given in Remark 29.

Definition 2 (Grammar).

$$\begin{aligned} P, Q &::= 0 \mid P \mid Q \mid \text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P \mid x!A \mid *_l x \mid *_r x \\ A, B &::= \langle \vec{P}, \vec{Q} \rangle \\ x, y &::= @_l A \mid @_r A \end{aligned}$$

Three points of reading. A send carries a pair of *sequences* of processes, not a process. A receipt binds a pair of sequences of name variables, of the same length, one sequence per polarity — this is forced, and Remark 11 gives the reason. And a name is a quotation of such a pair, tagged with a sector l or r .

Definition 3 (Rank, and well-formedness). SETTLED. A pair $\langle \vec{P}, \vec{Q} \rangle$ is well formed when $|\vec{P}| = |\vec{Q}|$, and its *rank* is that common length; the rank of a name is the rank of the pair it quotes, and the rank of a prefix is the common length of its two binder sequences, or of its two payload sequences. A receipt $\text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P$ is well formed when $|\vec{y}_p| = |\vec{y}_n|$. Only well-formed terms are terms.

Remark 4 (Why the two sequences must have equal length). $\ominus$ exchanges the two sequences (Definition 8), so it preserves rank only under the condition of Definition 3. Since x and $\ominus x$ must share an address (Definition 18) and rank is part of an address (Definition 26), unequal lengths would put a name and its negation at different addresses and the polarity structure would not close. The condition is not a patch: a name is a formal difference of two elements of the *same* grade, which is the only difference a graded module admits.

Remark 5 (Why pairs of sequences and not sequences of pairs). SETTLED. The sign is one $\mathbb{Z}/4$ element per name, global to it, not one per slot. Were each slot to carry its own polarity, $\ominus$ would negate every slot and so act as $(-1)^n$ on a name of rank n — the identity on even ranks, hence $\ominus(A \otimes B) = A \otimes B$, which is trivial exactly where Section 7 makes it interesting. Designating one privileged slot instead is not canonical and does not survive concatenation.

Convention 6 (Drops in process position). $*_s x$ is stipulated above to be a process, whereas the pair quoted by x has two components. We take $*_s x$ to denote the *positive* component of the s -sector content of x , so that $*_s(\ominus x)$ is the negative component. No generality is lost and the ordinary reading of $*$ is preserved.

Convention 7 (Abbreviations). We write $x!Q$ for $x!\langle Q, 0 \rangle$. Together with Convention 6 and the choice $@ := @_l$, $* := *_l$, this is what makes the ordinary rho calculus a sub-language; see Section 8.

3.2 Negation

Definition 8 (Negation on pairs and names). $\ominus$ is defined on pairs and on names by component exchange at the top:

$$\ominus \langle P, Q \rangle := \langle Q, P \rangle, \quad \ominus @_s A := @_s(\ominus A) \quad (s \in \{l, r\}).$$

The definition does not recurse into the components.

Proposition 9 (Negation is a total involution on names). $\ominus$ is defined on every name without reference to reduction, and $\ominus \ominus x = x$.

Proof. Immediate from component exchange being an involution on ordered pairs. $\square$

Proposition 10 (Unique lifting to processes). *SETTLED.* There is exactly one extension of $\ominus$ from names to processes which is a homomorphism of the term algebra, namely

$$\begin{aligned} \ominus 0 &:= 0, & \ominus(P \mid Q) &:= \ominus P \mid \ominus Q, & \ominus(x! \langle P, Q \rangle) &:= (\ominus x)! \langle \ominus P, \ominus Q \rangle, \\ \ominus(\text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x) P) &:= \text{for}(\langle \vec{y}_n, \vec{y}_p \rangle \leftarrow \ominus x) \ominus P, & \ominus(*_s x) &:= *_s(\ominus x). \end{aligned}$$

It satisfies $\ominus \ominus P = P$, and it descends to $\equiv$ and to $\equiv_N$.

Proof. Processes are freely generated by the term formers over names, so a map on names extends uniquely to an endomorphism of the term algebra; the displayed clauses are that extension. Involutivity follows by induction from Proposition 9. For $\equiv$, the clauses are homomorphic and leave the binder alone, so associativity, commutativity and the unit are preserved. For $\equiv_N$, applying $\ominus$ to the quote-drop law (Definition 24) at x yields the quote-drop law at $\ominus x$, since $*_s(\ominus x)$ is the negative component of x . $\square$

Remark 11 (Why the binder must be a pair of sequences). *SETTLED*, and it corrects an error of version 2, which bound a single sequence. The clause for the receipt in Proposition 10 *exchanges* the two binder sequences; it does not push $\ominus$ inside. That is not a stylistic choice, and the single-binder presentation cannot be repaired.

$\ominus$ is defined by structural recursion over names, and names are quotations (Definition 2); there is no clause for a name *variable*, because variables are not names but placeholders for them. So $\ominus y$ is not a term. With a single binder sequence, $\ominus$ applied to a receipt would produce a body containing $\ominus y_i$ for bound y_i , and the lift would be undefined on open terms — which is fatal, since every body under a binder is open. With two binder sequences, $\ominus$ never has to be applied to a variable: it permutes which variable plays which role, and the terms it produces are terms.

This also withdraws the argument by which version 2 removed the second binder. That argument said the negative variable is redundant because it equals $\ominus$ of the positive one — which presupposes $\ominus$ on variables, the very thing that is unavailable. The redundancy claim was circular. Making $\ominus$ a primitive former on names would be the alternative repair, at the price of names that are no longer simply quotations; it is not taken.

Remark 12 (What the lift is, and is not). It is worth being explicit, because the notation invites a conflation that the rest of the note depends on avoiding. The lifted $\ominus$ is an *automorphism of the term algebra* — a relabelling of the namespace — and not an additive inverse. In particular $P \mid \ominus P$ is inert and is not 0 : a send and its mirror have no receipt between them, so nothing reduces. Cancellation does not come from $\ominus$ on processes; it comes from the completion equations on pairs, and orthogonality accordingly comes from where Remark 74 says it does.

Homomorphism for $\mid$ is therefore a *consequence* of freeness, not a design decision, and it carries none of the algebraic content it would carry if $\ominus$ were an inverse.

Remark 13 (Why the pair is inside the quotation). *SETTLED.* If pairs occurred only on messages, then names would quote single processes and $\ominus x$ would be undefined: negation would be an operation on messages that the namespace cannot express, so x and $\ominus x$ would not both

be addresses, the polarity of a participant in a communication could not be stated, and Proposition 10 would have nothing to lift. Placing the pair inside the quotation makes negation an operation on addresses and lets the reduction rule mention $\ominus x$ as a name like any other.

There is a second reason, of a different kind. Under the reading in which parallel composition is addition on processes and name multiplication is $@P \cdot @Q = @(P \mid Q)$, quotation is an exponential. Completing to formal differences *before* exponentiating gives a namespace that is the exponential image of a group; completing afterwards does not, and that is why earlier attempts to obtain negation dynamically, as a consequence of annihilation, kept having to produce the sign at reduction time rather than having it available in the syntax.

4 Sectors, polarity, and the group of a name

4.1 Two quotations force two drops

Proposition 14 (Collapse). *Suppose the calculus has two quotation operators $@_l, @_r$ and a single dropping operator $*$ satisfying $*(@_s A) = A$ for both s , together with the quote-drop law $@_s(*x) \equiv_N x$. Then $@_l A \equiv_N @_r A$ for every A .*

Proof. Instantiate the quote-drop law at $x := @_r A$ and $s := l$: $@_l A = @_l(*(@_r A)) \equiv_N @_r A$. $\square$

So the two quotations are distinguishable only if the drops are split as well; a map has at most one two-sided inverse. This is why Definition 2 carries $*_l$ and $*_r$ and not a single $*$.

Definition 15 (Sector reading, by rotation). SETTLED. $*_s$ reads a name *into* the sector s , rotating it there first when it is not already there:

$$*_s(@_s A) := A, \quad *_l(@_r A) := A, \quad *_r(@_l A) := \ominus A.$$

Equivalently, $*_s x := *_s(i^{-k} x)$ where k is the least rotation of Definition 19 carrying x into sector s . The asymmetry between the two cross clauses is exactly the asymmetry of Definition 19: the descending passage from r to l carries the pair exchange, the ascending one does not.

Remark 16 (Rotation, not projection). SETTLED, replacing an earlier clause that sent the cross cases to $\langle 0, 0 \rangle$. Three reasons, of increasing weight.

Equivariance. Every other convention in this design has been fixed by requiring compatibility with the $\mathbb{Z}/4$ action of Definition 19; see Remark 31. Zero is a fixed point of that action, so a clause landing on zero cannot commute with it, and the drops would be the one place in the calculus where the group action is not respected. Rotation commutes with it by construction.

Reflection. With a projection to zero, $@_l(*_l(@_r A)) = @_l\langle 0, 0 \rangle \neq @_r A$, so the quote-drop law holds only when the sectors agree and reconstruction is available only on the diagonal. That is a weakening of the one property for which this calculus was chosen as a host. Under rotation the law is unconditional; see Definition 24.

Observability. Under projection, a context may apply both drops to a name and read off which one yields 0, so the sector is classically readable — a superselection rule rather than a phase, and in tension with Definition 26, which makes $@_l A$ and $@_r A$ one address. Under rotation both drops succeed and differ by a phase, so no context can read the sector by comparing them. Global phase is unobservable and relative phase is not, which is the correct behaviour, and it arrives as a consequence rather than as a discipline imposed by restricting where drops may occur.

Two costs, stated plainly. The drops become interdefinable — $*_r x = \ominus *_l (i^{-1}x)$ throughout — so they are no longer independent term formers, and Proposition 14 must be read with Remark 17 in view. And a drop may now introduce a sign into process position, which adds one more source to the occurrence counting of Obligation 14.

Remark 17 (The collapse argument still applies). Proposition 14 shows that a *single* drop forces the two quotations to coincide. Rotation does not supply a single drop; it supplies two drops related by an operation which is not itself a drop, so the argument is untouched in substance. It is no longer the clean statement it was when the drops were independent, and it should be reproved rather than assumed — see Obligation 10.

4.2 The four names of an address

Each name has an underlying pair and two binary attributes: its sector, l or r , and its polarity, which is the choice between a pair and its exchange. Fixing the underlying pair up to exchange therefore leaves four names.

Definition 18 (Address and phase). For $x = @_s A$, write $\text{sec}(x) := s$ and let $\text{addr}(x)$ be the pair A taken up to $\ominus$. The four names sharing an address are

$$@_l A, \quad @_r A, \quad @_l(\ominus A), \quad @_r(\ominus A).$$

Definition 19 (The imaginary unit). SETTLED, with the caveat of Remark 21. Multiplication by i is the successor map on the cycle

$$@_l A \mapsto @_r A \mapsto @_l(\ominus A) \mapsto @_r(\ominus A) \mapsto @_l A.$$

Equivalently: passing from l to r leaves the pair alone, and passing from r to l exchanges it.

Proposition 20. *The map of Definition 19 has order four, its square is $\ominus$, and the four names of an address form a torsor under $\mathbb{Z}/4$ with $\text{ph}(-)$ the resulting coordinate, $\text{ph}(@_l A) = 1$.*

Proof. Two applications carry $@_l A$ to $@_l(\ominus A)$, which is $\ominus$ by Definition 8; $\ominus$ is an involution by Proposition 9, so the map has order four. The action is free and transitive on the four names by Definition 18. $\square$

Remark 21 (Why i is not the composite of two independent involutions). This is the most delicate point in the design and it must not be glossed. Sector exchange and pair exchange are each involutions, and if they are taken as *independent* the group they generate is the Klein four-group. The Klein four-group is the group of the split-complex numbers, in which $j^2 = +1$ and there are zero divisors, $(1+j)(1-j) = 0$. Zero divisors are fatal here, since a pairing would then vanish for reasons having nothing to do with orthogonality. Definition 19 is precisely the stipulation that the two exchanges are *not* independent: the descending passage from r to l carries the pair exchange with it, so that the generated group is cyclic of order four rather than Klein. The asymmetry is not decoration. It is the content of the requirement $i^2 = -1$, and it is what a design with two free involutions cannot supply.

Remark 22 (Only one doubling). Name multiplication is parallel composition, hence commutative. The Cayley–Dickson construction preserves commutativity once and loses it at the second step. A namespace whose product is parallel composition therefore admits exactly one doubling and no more: the calculus can be complex, and cannot be quaternionic, for a structural reason rather than by stipulation.

5 Structural theory

Definition 23 (Structural congruence). $\equiv$ is the least congruence containing alpha-equivalence and satisfying associativity, commutativity and 0 as unit for $|$, together with

$$\ominus \ominus A \equiv A.$$

Definition 24 (Name equivalence). $\equiv_N$ is the least congruence satisfying

$$@_s(*_s x) \equiv_N i^k x, \quad \frac{A \equiv B}{@_s A \equiv_N @_s B}$$

where k is the rotation of Definition 15. The first law is unconditional: under rotation every name is reconstructible from either drop, up to the phase the rotation introduced, so reflection holds at every name and not only on the diagonal.

Remark 25 (The completion equations, and where they live). FORK, and the most consequential one in the note. A pair is intended as a formal difference, which suggests two further equations:

$$\langle Q, Q \rangle \equiv 0, \quad \langle P, Q \rangle \equiv \langle P \mid R, Q \mid R \rangle.$$

These are what make the pair sort the group completion of processes rather than merely an ordered pair. But $\equiv_N$ contains $\equiv$ under quotation, and COMM fires on $\equiv_N$. Imposing them therefore makes them equations *on addresses*, which the matcher must decide.

The two readings differ sharply.

- **Arithmetic names.** The completion equations hold on names. Cancellation is then real, $\ominus$ is genuine negation, and destructive interference is the matcher discovering that a pair's components have become congruent. The cost is that unique factorisation is lost, the monomial basis with it, and the decidability of name matching becomes a serious question.
- **Coloured names.** The completion equations do not hold on names, which are then a free set with a $\mathbb{Z}/4$ action. Negation is a label carried along and read by the reduction rule, exactly as the sector is. The matcher is untouched; the arithmetic is deferred to a coefficient layer outside the calculus.

The reduction rule of Section 6 never asks whether two differently-signed names are equal, only whether they share an address, so the coloured reading suffices for the operational semantics as it stands. The arithmetic reading is what the semantic programme wants. The note takes neither; it records that the choice determines whether cancellation is a structural event or an external one.

6 Reduction

6.1 Matching is sector-blind and polarity-blind

Definition 26 (Address matching). SETTLED. A receipt at x and a send at x' are *matched* when $\text{addr}(x) = \text{addr}(x')$, irrespective of $\text{sec}(x)$, $\text{sec}(x')$ and irrespective of polarity.

This is forced by the fourth defect of Section 2. If $@_l A$ and $@_r A$ were distinct addresses, then $@_l A!Q \mid @_r A!Q$ would be two messages that no single receiver can hear, rather than a superposition of one message with itself in two sectors; nothing would ever combine, the sector would be a superselection rule, and the doubling would carry no information. Sector and polarity are coefficients on a name, not part of its identity.

6.2 The rule

Definition 27 (Relative polarity). For matched x and x' , the *relative polarity* $\delta(x, x')$ is the identity when $x' = x$ and $\ominus$ when $x' = \ominus x$. It is well defined by Definition 18, since matched names differ by an element of the $\mathbb{Z}/4$ torsor and polarity is the component of that element which Definition 19 squares to.

Definition 28 (COMM). Let $t := \text{sec}(x)$ and let $\text{op}(l) := r$, $\text{op}(r) := l$. For matched x, x' in the sense of Definition 26,

$$\text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P \mid x'!(\vec{Q}_p, \vec{Q}_n) \rightarrow P\{N_i/y_{p,i}, \ominus N_i/y_{n,i}\}_i$$

where, writing $\delta := \delta(x, x')$ for the relative polarity and $u := \text{op}(t)$,

$$N_i := @_u(\delta\langle Q_{p,i}, Q_{n,i} \rangle).$$

The rule is closed under parallel composition and under $\equiv$ in the usual way.

One clause, two effects, and they remain independent of one another.

Sector: communication intertwines. SETTLED. The quotation deposited is in the sector *opposite* to the one the receipt listened on. A communication on the real axis deposits an imaginary name; a communication on the imaginary axis deposits a real one. So a communication event is the map $l \mapsto r \mapsto l$, and by Definition 19 it is multiplication by i .

Polarity: opposition exchanges the pair. SETTLED. When the participants have opposite polarity, δ is $\ominus$ and the deposited pair is exchanged, which by Definition 8 is multiplication by $\ominus$. So opposite polarity contributes the sign, and $i^2 = \ominus 1$ is realised as: two sector rotations equal one polarity exchange.

Remark 29 (One clause, two binder sequences). Version 1 of this note stated COMM in two clauses, one per relative polarity, and bound a pair of variables. Version 2 collapsed both, to one clause and one binder. Only the first collapse survives.

One clause stands. The polarity case is not a second rule but the factor δ , which is the statement that the relative group element of the two participants is the phase deposited — visible in the rule rather than distributed over two of them.

One binder does not. Version 2 argued that the negative variable is redundant, being $\ominus$ of the positive one. Remark 11 shows the argument is circular: $\ominus$ is not defined on variables, so the purported derivation of one binder from the other is not a term of the calculus, and with a single binder the lifting of $\ominus$ is undefined on open terms. The pair of sequences is restored, and what it buys is precisely that $\ominus$ acts on a receipt by *exchanging* the two sequences.

Version 1's deposit convention remains withdrawn, for the reason in Remark 31: it is the convention, not the binder, that equivariance rejects. Each variable now receives a rank-one name that is itself a formal difference, N_i or $\ominus N_i$, rather than a payload parked in the positive slot.

Proposition 30 (Equivariance). $P \rightarrow P'$ implies $\ominus P \rightarrow \ominus P'$.

Proof. By Proposition 10, $\ominus$ carries the redex $\text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P \mid x'!A$ to $\text{for}(\langle \vec{y}_n, \vec{y}_p \rangle \leftarrow \ominus x) \ominus P \mid (\ominus x')!(\ominus A)$, which is again a redex: matching is polarity-blind (Definition 26)

and $\delta(\ominus x, \ominus x') = \delta(x, x')$. Reducing it binds $y_{n,i}$ to $@_u(\delta\langle Q_{n,i}, Q_{p,i} \rangle) = \ominus N_i$ and $y_{p,i}$ to $\ominus \ominus N_i = N_i$, which is the image under $\ominus$ of the original substitution, since $\ominus$ leaves the N_i in the roles the exchanged binder assigns them. Hence both routes give $\ominus P\{N_i/y_{p,i}, \ominus N_i/y_{n,i}\}$, using $\ominus @_s B = @_s(\ominus B)$ and the commutation of $\ominus$ with substitution. $\square$

Remark 31 (Equivariance is a selection principle). Proposition 30 is not decoration; it is what picks the deposit convention out of the alternatives. Requiring reduction to commute with the unique lifting of Proposition 10 rules out the earlier convention, under which the deposited name was always $@_{\text{op}(t)}\langle Q, 0 \rangle$ with the payload in the positive slot: nothing of that shape lies in the image of an equivariant rule, since $\ominus @ \langle Q, 0 \rangle = @ \langle 0, Q \rangle$, and the failure is uniform at one application of $\ominus$ per communication. Depositing the whole pair is the unique repair, and it is the convention of Definition 28. That the same convention also collapses the second COMM clause is the reason to prefer it on grounds other than symmetry. It does *not* collapse the binder; see Remark 11.

Remark 32 (Phase is path length). An immediate consequence of the sector effect. The sector of a name is the parity of the number of communications that produced it, so the relative phase between two derivations reaching a common outcome is the parity of the difference of their lengths. This is worth naming, because the question of where destructive interference could come from in an unmodified ρ calculus had the answer *length-asymmetric reconvergence*, with no mechanism to supply it: concurrency diamonds are symmetric, so the holonomy vanishes. Here the phase *is* the length asymmetry, with no cost functor, no weight map and no external coefficient algebra.

Remark 33 (The identity step is not fictitious). Under the reading of Remark 32, a step that changes no process still rotates the sector. A device introduced elsewhere to complete deficient rows, and justified there case by case, acquires an independent justification here: doing nothing and doing nothing once are genuinely different.

6.3 The four-case refinement

Remark 34. FORK. Definition 28 computes the deposited sector from the receipt's channel alone. But Definition 26 makes matching sector-blind, so a receipt at $@_l A$ may be served by a send at $@_r A$, and the rule as stated ignores the sender's sector. The natural completion is that the two sectors *multiply*:

	send l	send r
recv l	1	i
recv r	i	-1

with -1 realised as $\ominus$ on the deposited pair and the off-diagonal entries being exactly the intertwining of Definition 28. Under this reading the rule of Definition 28 is the off-diagonal half of the table, and $i^2 = -1$ is a case of the reduction rule rather than a separate stipulation.

The refinement is attractive and is not adopted here, because it changes what the rule computes rather than extending it: on the diagonal it deposits a name in the sector it received on, which contradicts the plain reading of “communication intertwines”. Settling this is the first item of Section 11.

Remark 35 (Amplitude is multiplicity). Under Definition 26, the expression $@_l A!Q \mid @_r A!Q$ is a single address carrying one copy in each sector, and with the pair supplying signs this is $(1 + i)$ times a send at A . Generally, the coefficient of a message is the multiset of sectors and

polarities its copies occupy, so Gaussian-integer coefficients are realised as multiplicity under parallel composition, with no external coefficient algebra and no tagging. This is the payoff that motivated Definition 26, and it should be the first thing checked against any proposed revision of the rule.

7 Tensor and par

With sequences in hand there are two ways to combine two prefixes at two channels, and they are the two multiplicatives. Both keep the channels together and differ only in what happens to the binders: sharing a binder merges the two continuations' access to a single datum, which is context-merging; keeping the binders apart gives each continuation its own. The distinction is not available in a calculus with only $|$, and its availability here is what allows $\otimes$ and $\wp$ to be different operations rather than one degenerate one.

7.1 The name product

Definition 36 (Name product). For names of equal rank and equal sector,

$$@_s\langle \vec{R}, \vec{S} \rangle \cdot @_s\langle \vec{R}', \vec{S}' \rangle := @_s\langle \vec{R} \mid \vec{R}', \vec{S} \mid \vec{S}' \rangle$$

with $|$ taken slotwise. For unequal sectors the product carries the phases with it, by Definition 19.

Componentwise parallel composition of the contents is *addition* of formal differences, and under the reading in which quotation is an exponential (Remark 13) addition of contents is multiplication of names. So Definition 36 says that channels multiply, and that is the operation both multiplicatives use on the channel.

7.2 Par

Definition 37 (Par of prefixes).

$$\begin{aligned} \text{for}(\vec{Y}^1 \leftarrow x)P_1 \wp \text{for}(\vec{Y}^2 \leftarrow u)P_2 &:= \text{for}(\vec{Y} \leftarrow x \cdot u)(P_1 \wp P_2) \{ \vec{Y} / \vec{Y}^1, \vec{Y} / \vec{Y}^2 \} \\ x!A \wp u!B &:= (x \cdot u)!(A \wp B) \end{aligned}$$

where $\vec{Y}$ abbreviates a binder pair $\langle \vec{y}_p, \vec{y}_n \rangle$, the fresh $\vec{Y}$ has the common rank, and $A \wp B$ is slotwise. Substitution of one binder pair for another is polarity-respecting: positives for positives, negatives for negatives.

Remark 38 (Par is a comultiplication). The identification of $\vec{y}^1$ and $\vec{y}^2$ with one $\vec{y}$ is a sharing, and sharing a datum between two continuations is copying it. Copying a basis element is the Frobenius comultiplication — in the graphical calculus, the spider that fans one wire out to two. So Definition 37 builds *correlated* composites, which is a legitimate and wanted operation, and it is not a tensor: see Remark 41.

7.3 Tensor

Definition 39 (Tensor of prefixes).

$$\begin{aligned} \text{for}(\vec{Y}^1 \leftarrow x)P_1 \otimes \text{for}(\vec{Y}^2 \leftarrow u)P_2 &:= \text{for}(\vec{Y}^1 ++ \vec{Y}^2 \leftarrow x \cdot u)(P_1 \otimes P_2) \\ x!A \otimes u!B &:= (x \cdot u)!(A ++ B) \end{aligned}$$

where $++$ is concatenation, taken slotwise on pairs, so that binder pairs concatenate polarity by polarity. Both are extended by

$$(P_1 \mid P_2) \otimes Q := (P_1 \otimes Q) \mid (P_2 \otimes Q), \quad 0 \otimes Q := 0$$

and symmetrically in the second argument.

Proposition 40 (Interchange). *For x, u of the appropriate ranks,*

$$(\text{for}(\vec{Y}^1 \leftarrow x)P_1 \otimes \text{for}(\vec{Y}^2 \leftarrow u)P_2) \mid (x!A \otimes u!B) \rightarrow P_1\{A/\vec{Y}^1\} \otimes P_2\{B/\vec{Y}^2\}$$

modulo the phase of Definition 28, provided substitution distributes over $\otimes$.

Remark 41 (Why the tensor needs separate binders). Under Definition 37 the same computation gives $P_1\{@(A \wp B)/\vec{Y}^1\} \wp P_2\{@(A \wp B)/\vec{Y}^2\}$: each continuation receives the *whole* combined payload rather than its own factor. Recovering the factors would require splitting a product of names, hence unique factorisation, which the completion equations of Remark 25 destroy. So interchange is exactly what separates the two operations, and it is the reason the tensor concatenates rather than shares.

Remark 42 (Rank is the tensor grade). Concatenation adds rank, so $\otimes$ grades the namespace, and the grading is free rather than symmetric because concatenation is not commutative. The symmetry of $\otimes$ is therefore an explicit permutation — a braiding, not strict commutativity — and symmetrisation or antisymmetrisation of a rank is a choice of twist rather than a property of the syntax. This does not reopen the Cayley–Dickson bound recorded above: that concerns the scalars, which remain commutative, so the calculus is still complex and not quaternionic.

7.4 Two consequences and two conjectures

Proposition 43 (Slotwise par is not the addition). *Let $A = \langle \vec{a}, \cdot \rangle$ and B be of rank two and C of rank one, and suppose addition of same-rank names were slotwise $\mid$. Then*

$$(A + B) \otimes C = \langle a_1 \mid b_1, a_2 \mid b_2, c \rangle, \quad (A \otimes C) + (B \otimes C) = \langle a_1 \mid b_1, a_2 \mid b_2, c \mid c \rangle$$

which differ: the untouched factor is duplicated. Hence $\otimes$ is not bilinear for slotwise $\mid$.

Corollary 44 (The sum in $\odot$ is external). *Remark 61 is resolved against the internal reading. The linear structure must be formal sums over names as basis elements; slotwise parallel composition retains a role, but as the name product of Definition 36, not as the addition. Names accordingly carry two operations, $\cdot$ and $\otimes$, and $\otimes$ does not distribute over $\cdot$ — which is expected of a graded product and is not a defect.*

Remark 45 (Zero, and the missing unit). $0 \otimes Q = 0$ makes 0 the additive unit and the multiplicative annihilator at once, so $0 \cdot P = 0$ holds: the unit collision of Section 2 is resolved at this end. The other end is not. $\otimes$ has no unit, since a unit would need channel content of rank zero and a body that is itself a unit, and no term serves for both prefix kinds. The structure is a non-unital graded rig, which is the same shape the join exhibits, and the fictitious identity step is its formal unitalisation.

Conjecture 46 (Multiplicativity of evaluation). $\Downarrow (P \otimes Q) = \Downarrow P \otimes \Downarrow Q$, equivalently

$$\langle P_1 \otimes P_2 \mid Q_1 \otimes Q_2 \rangle = \langle P_1 \mid Q_1 \rangle \cdot \langle P_2 \mid Q_2 \rangle.$$

This is the criterion by which the operation earns its name, and it is what makes the failure of $\otimes$ to preserve single steps harmless. By distributivity a redex of P appears once per component of Q in $P \otimes Q$, so one step becomes a layer of steps and the image of $\otimes$ is not closed under $\rightarrow$; but the pairing looks only at terminal states, so non-closure in between costs nothing provided the endpoints agree. Two things follow if the conjecture holds: $(P \otimes Q)^\dagger = P^\dagger \otimes Q^\dagger$, and the states of Definition 68 are closed under $\otimes$, which is what makes it an operation on states rather than on terms.

Conjecture 47 (De Morgan). $(P \otimes Q)^\dagger = P^\dagger \wp Q^\dagger$.

If this holds the structure is $*$ -autonomous rather than degenerately compact-closed, and the operator algebra ceases to be weaker than the state space. It is the criterion the two multiplicatives must jointly satisfy, and it is the reason for defining both rather than only the one the semantics immediately needs.

7.5 The heterogeneous cases

Remark 48 (Mixed prefix kinds are not defined, and should not be). SETTLED. $\otimes$ and $\wp$ are defined on like prefixes — receipt with receipt, send with send — and are undefined on a receipt against a send. That is not a gap. A receipt and a send at a common address are the two poles of an interaction, and their combination is the *pairing* of Definition 72, not a monoidal product: in the graphical calculus $\otimes$ is the product and the mixed case is the cup. The tensor-of-agents material in the earlier work restricts to what it calls homogeneous pairings for the same reason.

Remark 49 (Drops, and openness). $*_s x \otimes Q$ unfolds when x is a concrete quotation and is stuck when x is a bound variable, so $\otimes$ is total on closed terms and partial on open ones. This is the reflection obstacle in its usual place and is not removed by anything in this section.

Remark 50 (Different sectors, and a coherence argument). $@_l A \otimes @_r B$ multiplies channels of different sectors, hence multiplies their phases, hence uses precisely the table of Section 6.3. So the sector rule for the multiplicatives and the sector rule for COMM are one table, and the fork recorded there cannot be settled independently for the two. This is an internal coherence argument and is therefore stronger than the conservativity argument of Remark 53, which is about backward compatibility with a calculus this one does not claim to extend.

Remark 51 ($\otimes$ is not a congruence). After tensoring, the composite listens at $x \cdot u$ and no longer at x , so messages elsewhere in an ambient term addressed to x can no longer reach it. $\otimes$ builds a new system rather than composing systems in place. This is the honest successor to the renaming device used in the earlier work to keep the two sides of a tensor from interacting: separation is now compositional address arithmetic rather than hygiene, which is a real improvement. It is still not *secure*, since $x \cdot u$ is computable by anyone; unforgeable separation continues to require a name-restriction operator.

8 Conservativity

Proposition 52 (Embedding). *FORK, stated as a target rather than a theorem. Let $\iota(-)$ send an ordinary rho term to the term of Definition 2 obtained by*

$$\begin{aligned} \iota(x!Q) &:= \iota(x)!\langle \iota(Q), 0 \rangle, & \iota(@P) &:= @_l \langle \iota(P), 0 \rangle, \\ \iota(\text{for}(y \leftarrow x)P) &:= \text{for}(\langle y, y' \rangle \leftarrow \iota(x))\iota(P), & \iota(*x) &:= *_l \iota(x) \end{aligned}$$

with y' fresh, homomorphically elsewhere. Then $\iota(-)$ is injective and preserves $\equiv$.

Remark 53 (Conservativity fails for reduction, and deliberately). The image of $\iota(-)$ is not closed under $\rightarrow$. A single communication moves a name off the real axis, so an ordinary ρ program is not, in this calculus, a program that stays classical. This is not an oversight to be repaired: it is the content of Remark 32, since a calculus in which ordinary programs remained on the real axis would be one in which no phase ever accumulated. The design is a genuinely different calculus, not an extension, and it should be presented as such.

If a conservative reduction is wanted, the diagonal entries of the table in Section 6.3 supply it: under that reading, a real receipt served by a real send stays real, and the ordinary calculus embeds as the real sub-calculus. That is an argument for the refinement, and is recorded here rather than in Section 6.3 because it is the strongest one.

9 Evaluation and residuation

The calculus above is a term language and a reduction relation. The construction it serves needs two further operations: an evaluation, which runs a composite to its terminal states and collects them, and a pole, against which the dagger is obtained by residuation. Both are recorded here. Following the design decision that the pole be taken as simply as possible for the time being, $\perp$ is the zero process and $P^\perp$ is residuation to it — the linear perp.

9.1 Inert states and evaluation

Definition 54 (Inert). P is *inert* when $P \not\rightarrow$; that is, when no COMM applies to any of its components under $\equiv$. Write **Inert** for the inert processes.

Inertness is not triviality: 0 is inert, and so is $x!A$ with no matching receipt. The distinction matters throughout what follows, and is the reason $\perp$ below is not simply “is inert”.

Definition 55 (Evaluation product).

$$P \odot Q := \sum_{\substack{R \in \text{Inert} \\ P|Q \rightarrow^* R}} R$$

where the sum ranges over terminal states with multiplicity, one summand per *causal history* ending at R .

Proposition 56 (Commutativity). $P \odot Q \equiv Q \odot P$.

Proof. $|$ is commutative and the index of the sum mentions P and Q only through $P|Q$. $\square$

Definition 57 (Evaluation). $\Downarrow P := 0 \odot P$.

Remark 58 (Histories, not paths). SETTLED, and load-bearing rather than hygienic. Indexing the sum by reduction *sequences* rather than by causal histories counts the linearisations of a partial order, so m independent redexes contribute a spurious factor of $m!$. The multiplicity in Definition 55 is therefore the number of histories, equivalently the number of maximal chains in the reduction order modulo permutation of independent steps. Concurrency squares are filled; what is summed is what survives the quotient.

Remark 59 (What $\odot$ is not). Three properties fail, and each failure is informative rather than repairable.

No unit. $0 \odot P = \Downarrow P$, which is P only when P is already inert. So 0 is a unit on **Inert** and nowhere else, which is one reason the carrier of the eventual algebra must be cut down — see Section 9.2.

Not idempotent. $P \odot P \neq P$ in general, because two copies of a process interact. This is a feature: it is what makes multiplicity meaningful, and it is why the additive structure cannot be mixed choice, whose idempotence would force $2 = 1$.

Not associative. $(P \odot Q) \odot S$ forces $P \mid Q$ to reach a terminal state before S arrives, so any global reduction of $P \mid Q \mid S$ that schedules a communication between P and S early, destroying a P – Q redex, is a history of the unbracketed composite and of no bracketed one. The bracketed product undercounts, and the deficit is exactly the cross-redexes. Writing M_R for the operator $\square \mid R$ and T for one-step reduction, the deficit is the commutator $[T, M_R]$: the redexes between R and the state that existed in neither alone. M_R alone is diagonal in the occupation basis and commutes with every M_S ; T alone is the off-diagonal part; neither is interesting, and their failure to commute is where the content of the construction lives.

Convention 60 (The n -ary form). Because of the last item, the primitive is not the binary product but the family of symmetric n -ary evaluations

$$\odot(R_1, \dots, R_n) := \sum_{\substack{R \in \mathbf{Inert} \\ R_1 \mid \dots \mid R_n \rightarrow^* R}} R$$

with one global evaluation and no bracketing. Symmetry under permutation is Proposition 56. Nothing needs to be associative because nothing is composed twice; factorisation of an n -ary evaluation into lower ones then becomes a property that holds exactly on causally independent components, rather than an axiom.

Remark 61 (What the sum is). **FORK**. Under the reading in which parallel composition is addition, the formal sum of Definition 55 is available in the calculus: it is the parallel composition of the terminal states. That reading is attractive, since it needs no new former and no tagging, and multiplicity is then counted by $\mid$ itself.

It has a hazard that must be checked before it is adopted. Inertness is not preserved by parallel composition: $x!A$ and $\text{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P$ are each inert and their composition is not. So the parallel composition of the summands may reduce further, and the sum would not be a normal form. The alternative is an external formal sum over the multiset of terminal states, which is inert by fiat and costs a coefficient layer outside the calculus.

Two things are settled regardless. Tagging the summands with fresh names is *not* the way to defeat idempotence: a retained tag is a which-path record, distinguishing summands permanently and preventing exactly the coincidence of outcomes on which the addition of coefficients depends. And quotation carries $\mid$ to the name product, so quoting an internalised sum yields a *product* of quotes; whichever reading is taken, the sum must not be internalised beneath a quotation.

Remark 62 (Partiality). $\odot$ is partial and the sum may be infinite. Non-confluence is not itself an obstacle — the sum is over all terminal states, not over a chosen one — but divergence is: a composite with no terminal state contributes the empty sum, and a composite with infinitely many contributes a sum that requires a summability condition before it denotes anything. Both are recorded as Obligation 2.

9.2 The pole, and residuation

Definition 63 (Residual). For formulae φ, ψ of the generated spatial logic, $\varphi \triangleright \psi$ is the right adjoint to the separating conjunction:

$$P \models \varphi \triangleright \psi \quad \text{iff} \quad \forall R. R \models \varphi \Rightarrow P \mid R \models \psi.$$

Definition 64 (The pole). $\perp$ is the behavioural formula

$$P \models \perp \quad \text{iff} \quad \Downarrow P \equiv 0,$$

that is, every terminal state of P is the zero process.

Definition 65 (Linear perp). $P^\perp := \{Q : P \odot Q \equiv 0\}$, and for a set S of processes, $S^\perp := \bigcap_{P \in S} P^\perp$.

Proposition 66 ($P^\perp$ is residuation to $\perp$). $Q \in P^\perp$ iff $Q \models \psi_P \triangleright \perp$, where ψ_P is a formula whose extension is the equivalence class of P in the logic.

Proof. By Definition 63, $Q \models \psi_P \triangleright \perp$ iff $Q \mid R \models \perp$ for every R in the extension of ψ_P ; by Definition 64 that is $\Downarrow (Q \mid R) \equiv 0$, which is $R \odot Q \equiv 0$ by Definitions 55 and 57. $\square$

Corollary 67 (Symmetry). $Q \in P^\perp$ iff $P \in Q^\perp$.

Definition 68 (States). P is a *state* when $\{P\}^{\perp\perp} = \{P\}^{\perp\perp}$ is generated by P , that is, when P is $\perp\perp$ -closed. The states are the carrier of everything that follows.

Remark 69 (The closure is the null-space quotient). $\{P\}^{\perp\perp}$ is the set of processes annihilated by every annihilator of P , so biorthogonal closure is closure under the testing preorder induced by the pole, with the tests being annihilators and success being reduction to the zero process. Three consequences.

It supplies the collapse that quotation cannot. Quotation is injective on $\text{Proc}/\equiv$ and therefore identifies nothing; every identification in the eventual pairing comes from $\odot$ and from this closure, that is, from reduction.

It fixes the equivalence at which the construction lives, and the equivalence it fixes is *testing*, not bisimulation. Since $\equiv_N$ is intensional and testing is not, the coefficient object and the state space sit at different equivalences; that is a known tension and it is not resolved here.

It destroys freeness. $\text{Proc}/\equiv$ under $\mid$ is free commutative on the non-parallel components, and closed sets are not freely generated. The occupation basis is a coordinate system for the unclosed calculus and does not survive into the closed one.

Remark 70 (Totality fails, and how it is repaired). Not every process has an annihilator: a persistent or replicating process at a name no one else holds has none, so $\psi_P \triangleright \perp$ is unsatisfiable and the dagger is partial. Definition 68 is the repair adopted here — restrict to the closed elements, which is the standard move — and it is the reason the states are cut down rather than the pole loosened. The alternative, kept in reserve, is to relativise the pole to a residue, taking $R \vdash P \perp Q$ when every terminal state of $P \mid Q$ is R ; that would make the pairing residue-valued in exactly the way the intended semantics wants, at the cost of a family of poles rather than one.

Remark 71 (Canonical witnesses are forced). $P^\perp$ is a set, and the dagger must be a term, because it is going to be quoted. Quotation does not respect any behavioural equivalence, so a representative must be chosen and not merely existentially quantified. The choice adopted is by minimal structural complexity, the measure already present for names and processes, with ties broken by the summation of Definition 55 when the minimum is not unique.

9.3 The pairing

Definition 72 (Dagger and pairing). For a state P , $P^\dagger$ is the canonical witness of $\psi_P \triangleright \perp$ in the sense of Remark 71, composed with the sector and polarity correction of Section 10. The pairing is

$$\langle P \mid Q \rangle := @_l \langle P^\dagger \odot Q, 0 \rangle$$

and, for a reified operator, using Convention 60,

$$\langle P \mid K \mid Q \rangle := @_l \langle \odot \left(P^\dagger, K_x^*, x! \langle Q, 0 \rangle \right), 0 \rangle$$

with x private to the composite and $K_x^* := \text{for}(\langle y, y' \rangle \leftarrow x) K[*_l y]$ the abstraction corresponding to K .

Remark 73 (Why the quotation is outermost, and why the operator is reified). Two points, each answering an earlier error.

$\odot$ lands in *Proc* and the scalars are names, so the pairing must pass through a quotation to change sort; a definition that has $\odot$ produce a scalar directly collapses the two sorts at precisely the place where the collapse is dangerous. The quotation is applied once, after evaluation, and contributes no identifications of its own.

Reifying K removes the bracketing question rather than answering it. Filling a hole and then evaluating, or evaluating and then filling, differ by the cross-redexes of Remark 59; passing the argument along a private wire means there is no hole-filling anywhere, and the bracket is manifestly the three-point evaluation it should have been. Privacy of x is required, not cosmetic: it is what prevents $P^\dagger$ from intercepting the operator’s wire, and renaming cannot supply it in a calculus where any party may construct any quotation.

Remark 74 (Annihilation is normalisation, not orthogonality). An immediate and initially unwelcome consequence. $\langle P \mid P \rangle = @_l \langle 0, 0 \rangle$, the unit of the name product, so annihilation gives 1. The pole is a *normalisation* condition: $P^\dagger$ is the process that reduces P to the unit, and $\langle P \mid P \rangle = 1$ is a theorem rather than a stipulation. That is a gain, but it vacates the position orthogonality was expected to occupy, and the correspondence table of the earlier work, which reads “orthogonality is process annihilation”, is backwards on this point.

Orthogonality must therefore come from cancellation among the summands of Definition 55, which requires additive inverses on those summands. This is precisely what the pair sort supplies and what the unexpanded namespace of Section 2 could not: two histories reaching the same terminal state with opposite polarity cancel, and $\langle P \mid Q \rangle = 0$ is the vanishing of a sum of nonzero contributions rather than the absence of any. The expansion of the namespace and the definition of the pairing are answerable to each other at exactly this point.

10 The dagger, in one paragraph

The construction this calculus serves requires an antilinear involution on states. Section 9 supplies the pole and the residuation; what remains is the sector and polarity content, recorded here.

First, the dagger is not a syntactic polarity swap on term formers. It is obtained by residuation against a pole: $Q^\dagger$ is a witness of $\psi_Q \triangleright \perp$, in the sense made precise by Proposition 66 and Remark 71. This makes the dagger a property first and a term second, hence stable under

any presentation generating the same logic. The choice of $\perp$ as the zero process is the simplest available and is provisional; Remark 70 records the relativised alternative.

Second, the sector and polarity components of the dagger are now fixed. Flipping both sectors is the map $(a, b) \mapsto (b, a)$, which in the complex numbers is $z \mapsto i \cdot \bar{z}$ and not conjugation; taken as the dagger it makes the pairing skew-Hermitian and puts $\langle P \mid P \rangle$ in the imaginary sector, so positivity fails outright. The correction is to compose the sector flip with a pair exchange,

$$@_l \mapsto \ominus @_r, \quad @_r \mapsto \ominus @_l,$$

which is $z \mapsto -i\bar{z}$, is involutive, and is assembled entirely from operations Definitions 8 and 19 already provide. Since Proposition 10 lifts $\ominus$ to processes, this composite acts on a whole process rather than name by name, which is what the pairing of Definition 72 requires and what an earlier presentation could not write down. Remark 12 remains in force: $\ominus$ supplies the dagger's polarity component and is not itself the dagger.

Remark 75 (The positivity obligation, made precise). Since every communication rotates the sector and the dagger rotates it once more, the sector in which $\langle P \mid P \rangle$ lands depends on the number of communications consumed in the annihilation of $P^\dagger$ against P , modulo four. Positivity therefore requires that all annihilation paths of $P^\dagger \mid P$ agree modulo four. This is a graded confluence condition, weaker than confluence, and it is not obviously true. It is cheap to test on the ladder of mutually annihilating names, and it should be tested before anything further is built on the dagger.

11 Obligations

Obligation 1 (Cross-redexes and the unit of evaluation). Remark 59 makes $\odot$ non-associative, with the deficit $[T, M_R]$. Convention 60 sidesteps the question for a single evaluation but does not answer it: composing operators, $\langle P \mid K_1 K_2 \mid Q \rangle$, reintroduces it in the form of whether the wires create cross-redexes. Determine the condition under which the n -ary evaluation factors, and check it against the reified operators of Definition 72.

Obligation 2 (Summability). Discharge Remark 62: give the condition under which the sum of Definition 55 denotes when the set of terminal states is infinite. Over a positive codomain monotone convergence suffices; with the pair sort supplying signs it does not, and an absolute-summability condition on the annihilation profile is the candidate.

Obligation 3 (The sum, internal or external). Settle Remark 61. If the sum is parallel composition, show that the composite of the terminal states is inert, or characterise when it is not. If external, say where the coefficient layer lives and how it interacts with Remark 25.

Obligation 4 (Matching the logic to the pole). Proposition 66 uses a formula ψ_P whose extension is the equivalence class of P . If that logic separates more finely than $\perp\!\!\!\perp$ -equivalence, then $P \mapsto P^\dagger$ does not factor through the states of Definition 68 and the dagger is ill defined on them. Either take the characteristic formula in a logic whose adequacy target is the pole's equivalence, or define the dagger on terms and prove separately that it descends.

Obligation 5 (The four-case table). Settle whether the deposited sector is computed from the receipt's channel alone (Definition 28) or from both participants (Section 6.3). Remark 53 is the strongest argument for the latter. This decision governs everything downstream and should be taken first.

Obligation 6 (Arithmetic or coloured names). Settle Remark 25: whether the completion equations hold on names, and hence whether cancellation is a structural event decided by the matcher or an external one deferred to a coefficient layer.

Obligation 7 (Decidability of name matching). COMM fires on $\equiv_N$. Under the arithmetic reading of Remark 25, the matcher must normalise formal differences nested through quotation. Termination is not obvious, and this is the practical gate on the entire design.

Obligation 8 (Freshness). The property that a quotation of P is not free in P is what obviates a freshness operator and underwrites freshness by quotation. Its proof proceeds by structural complexity. Both quotation operators must increase that measure, which is routine; but the quote-drop law of Definition 24 relates a name to a term built from its own drops, and the completion equations, if adopted, shorten terms non-structurally. The mutual recursion of $\equiv$ and $\equiv_N$ was already the delicate part of the metatheory. That argument must be redone rather than adapted.

Obligation 9 (Observability of the sector). DISCHARGED by Definition 15, and recorded because it was open in earlier versions. When the cross cases of the drops projected to 0, a context could apply both drops and read the sector, making it a superselection rule; the remedy contemplated was to restrict drops to guard position. Under rotation both drops succeed and differ by a phase, so the sector is not readable by comparing them and no restriction on where drops may occur is needed. See Remark 16.

Obligation 10 (Reprove the collapse proposition). Under Definition 15 the two drops are interdefinable, so Proposition 14 no longer concerns independent term formers. Remark 17 argues that the substance survives; the proof should be redone rather than assumed.

Obligation 11 (Substitution and the tensor). Proposition 40 is conditional on substitution distributing over $\otimes$. Discharge or refute it; the interchange law, and with it the distinction between the two multiplicatives, turns on this.

Obligation 12 (The two conjectures). Settle Conjectures 46 and 47. The first decides whether $\otimes$ deserves its name and whether the states are closed under it; the second decides whether the structure is *-autonomous or degenerately compact-closed.

Obligation 13 (Rank and arity). Definition 3 gives a name a rank and a prefix an arity, and Definitions 36 and 39 keep them apart: channels multiply, so their contents combine slotwise and their rank is unchanged, while payloads concatenate and arity adds. Whether the two numbers should instead be identified — which would require channels to concatenate as well, making $\otimes$ uniformly concatenation — is open, and it is the cleanest remaining question about the graded structure.

Obligation 14 (Scalar extraction and occurrence counts). For a race to interfere, its candidates must reach the same outcome with different coefficients. Consuming from $@_l A$ rather than $@_r A$ yields $P\{@_r Q/y\}$ rather than $P\{@_l Q/y\}$, and these are the same outcome up to phase only if the sector can be extracted from the name to the term, that is, only if there is a homomorphism from processes to $\mathbb{Z}/4$ under which substitution multiplies. The natural candidate is sector degree, the product of sectors over name occurrences — but then the phase is raised to the power of the number of occurrences of y , so extraction holds only when y occurs exactly once. Occurrence count is the exponent on the phase. Either linear use of names bound from graded channels

is required, or multiplicative phase in occurrences is accepted and scalar extraction abandoned. This is a linearity requirement of a different kind from the ones previously considered and declined, and the earlier reasons for declining do not apply to it.

Obligation 15 (Positivity). Discharge or refute Remark 75: that all annihilation paths of $P^\dagger \mid P$ agree in length modulo four.

Obligation 16 (Two gradings on names). This calculus grades names by sector. A separate line of work grades them by binding structure, unforgeable names being those introduced by a name-restriction operator and classical ones being quotations. The two gradings are independent, so there are four kinds of name, and their interaction is unstated. The conjecture worth testing is that sectors should live only in the unforgeable namespace, which would also confine Obligation 9.

Obligation 17 (Metatheory). Nothing in this note is proved beyond Propositions 9, 14 and 20. Subject reduction for the sector grading, whether bisimulation is a congruence in the presence of two quotations, and whether the sector is visible to bisimulation at all, are all open. The last is pressing: an inert distinction that $\equiv$ can see and $\sim$ cannot is exactly the configuration in which the spatial layer strictly refines behaviour, and the sign structure would then be behaviourally invisible.

12 What this note does not do

It does not give the interpretation map, and it does not establish that the resulting scalars form a rig — only that the four obstructions to their doing so have each been given a place to be repaired. The evaluation and the pole of Section 9 are stated, not developed: nothing is proved about $\odot$ beyond commutativity, the pole is the simplest one available and is provisional, and the summability question of Obligation 2 is open, so the pairing of Definition 72 is a definition and not yet a well-defined operation. It does not treat the join, whose generalisation to graded messages requires a sum over matchings with coefficients in a group ring, and whose normalisation may or may not remain the right unit. And it takes no position on whether the resulting calculus is worth the cost of Obligation 7, which is the question a reader should keep in view throughout.

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